

The effect of Mediterranean diet on metabolic syndrome and its components: a meta-analysis of 50 studies and 534,906 individuals.

OBJECTIVES: The aim of this study was to meta-analyze epidemiological studies and clinical trials that have assessed the effect of a Mediterranean diet on metabolic syndrome (MS) as well as its components.

BACKGROUND: The Mediterranean diet has long been associated with low cardiovascular disease risk in adult population.

METHODS: The authors conducted a systematic review and random effects meta-analysis of epidemiological studies and randomized controlled trials, including English-language publications in PubMed, Embase, Web of Science, and the Cochrane Central Register of Controlled Trials until April 30, 2010; 50 original research studies (35 clinical trials, 2 prospective and 13 cross-sectional), with 534,906 participants, were included in the analysis.

RESULTS: The combined effect of prospective studies and clinical trials showed that adherence to the Mediterranean diet was associated with reduced risk of MS (log hazard ratio: -0.69, 95% confidence interval [CI]: -1.24 to -1.16). Additionally, results from clinical studies (mean difference, 95% CI) revealed the protective role of the Mediterranean diet on components of MS, like waist circumference (-0.42 cm, 95% CI: -0.82 to -0.02), high-density lipoprotein cholesterol (1.17 mg/dl, 95% CI: 0.38 to 1.96), triglycerides (-6.14 mg/dl, 95% CI: -10.35 to -1.93), systolic (-2.35 mm Hg, 95% CI: -3.51 to -1.18) and diastolic blood pressure (-1.58 mm Hg, 95% CI: -2.02 to -1.13), and glucose (-3.89 mg/dl, 95% CI: -5.84 to -1.95), whereas results from epidemiological studies also confirmed those of clinical trials.

CONCLUSIONS: These results are of considerable public health importance, because this dietary pattern can be easily adopted by all population groups and various cultures and cost-effectively serve for primary and secondary prevention of the MS and its individual components. Copyright © 2011 American College of Cardiology Foundation. Published by Elsevier Inc. All rights reserved.